

Section 25: STP

Layer: L2 | Reference: Odom Vol1 Ch9-10: STP, RSTP and EtherChannel | *Original study notes — CCNA 200-301*

Spanning Tree Protocol (STP) prevents Layer 2 loops in switched networks with redundant paths. Without STP, a redundant topology would create broadcast storms and MAC table instability within seconds.

Key Concepts

- STP's job: build a loop-free logical topology over a physically looped network by blocking redundant links
- Root Bridge election: lowest Bridge ID (BID) wins. BID = Priority (default 32768) + MAC address
- To manually force a switch to become root: lower its priority (spanning-tree vlan <id> root primary, or set priority directly)
- Every non-root switch selects one Root Port: the port with the lowest cost path to the root bridge
- Path cost is cumulative and based on link speed: 10 Mbps=100, 100 Mbps=19, 1 Gbps=4, 10 Gbps=2 (IEEE short path cost values)
- Designated Port: the forwarding port on each network segment, chosen by lowest cost to root, then lowest sender BID, then lowest sender port ID
- Blocking ports: any port that is neither root nor designated; this is what prevents the loop
- Original 802.1D STP port states: Blocking -> Listening -> Learning -> Forwarding (each transition takes ~15-20s by default, total convergence ~30-50s)
- RSTP (802.1w, Rapid STP) reduces port states to Discarding, Learning, Forwarding and converges in seconds using proposal/agreement instead of timers
- PortFast immediately puts an access port into forwarding state, skipping listening/learning - use only on ports connecting to end devices, never to other switches
- BPDU Guard: err-disables a PortFast port if it receives a BPDU, protecting against accidental switch loops on access ports
- Root Guard: prevents a port from becoming a root port, protecting against a rogue switch claiming to be root
- Loop Guard: prevents a non-designated port from becoming designated and forwarding when BPDUs stop being received, guarding against unidirectional link failures
- PVST+ runs a separate STP instance per VLAN (Cisco default); MST (Multiple Spanning Tree) groups VLANs into instances for scalability

Command Reference

Command	Purpose
show spanning-tree	Display STP status per VLAN: root bridge, port roles, states, costs
show spanning-tree vlan <id>	Show STP detail for a specific VLAN
spanning-tree vlan <id> root primary	Lower this switch's priority enough to become root for the VLAN

Command	Purpose
<code>spanning-tree vlan <id> priority <value></code>	Manually set bridge priority (must be a multiple of 4096)
<code>spanning-tree portfast</code>	Enable immediate forwarding on an access port (interface config mode)
<code>spanning-tree bpduguard enable</code>	Err-disable the port if a BPDU is received - pairs with PortFast
<code>spanning-tree guard root</code>	Prevent a port from ever becoming a root port
<code>spanning-tree mode {pvst rapid-pvst mst}</code>	Select the STP variant running on the switch

Configuration Walkthrough

Force SW1 to be root for VLAN 10, enable Rapid PVST

```
SW1(config)#spanning-tree mode rapid-pvst
SW1(config)#spanning-tree vlan 10 root primary
```

Configure PortFast + BPDU Guard on an access port

```
SW1(config)#interface GigabitEthernet0/5
SW1(config-if)#switchport mode access
SW1(config-if)#spanning-tree portfast
SW1(config-if)#spanning-tree bpduguard enable
```

Worked Scenario

Q: A switch port connected to an end-user PC suddenly goes err-disabled right after you plugged in a small unmanaged switch instead. Why?

BPDU Guard fired. The port had PortFast + BPDU Guard enabled (correct for an access port), but a BPDU reached the port - which PortFast ports should never see. Either remove the rogue device causing an accidental loop back into the network, or, if a switch genuinely belongs there, move that port off PortFast and treat it as a normal trunk/access port participating fully in STP.

Common Mistakes / Gotchas

- Enabling PortFast on a port that connects to another switch - this risks a transient loop before STP can react
- Forgetting that lower priority numbers win the root election (counterintuitive to many)
- Confusing root port (per switch, toward root) with designated port (per segment, forwarding)
- Not realizing each VLAN can have a different root bridge under PVST+

Exam Tips

★ *BID = Priority + MAC. Lowest BID = root. This is foundational and tested repeatedly.*

- ★ *Know the RSTP port states (Discarding/Learning/Forwarding) vs classic STP (Blocking/Listening/Learning/Forwarding)*
- ★ *PortFast + BPDU Guard is a classic exam pairing for access-port best practice*